

Experimental Determination of Planck's Constant

Using LED Threshold Voltage and Wavelength

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Research Question

How can Planck's constant (h) be determined by investigating the relationship between the threshold voltage (V) and wavelength (λ) of light emitted by different colors of light emitting diodes?

Background information

In the late 19th century, physicists was facing the challenge of being unable to explain the blackbody radiation problem proposed by Kirchhoff. A blackbody, an idealized object that absorbs and emits all radiation perfectly, was found to emit a spectrum of radiation depended only on its temperature. Two classical theories that attempted to explain the phenomenon—Wien's law and the Rayleigh-Jeans law—failed to account for all wavelength ranges. The former performed well at short wavelengths but severely underestimated radiation energy at long wavelengths. The latter accurately described radiation at long wavelengths but predicted an infinite energy density at short wavelengths, leading to the so-called “ultraviolet catastrophe” (Mavani & Singh, 2024).

It wasn't until 1900 that Max Planck, while trying to derive a mathematical expression that could reconcile both theories, proposed a revolutionary idea that solved the problem: energy is quantized (Britannica, 2024). He hypothesized that electromagnetic energy could only be emitted or absorbed in discrete packets called “quanta”, which were later called photons, with energy proportional to their frequency. This is now known as the Planck-Einstein relation (Briceño, 2019):

$$E = h \times f \quad (1)$$

where:

- E - Energy of a photon (J)
- h - Planck's constant (Js)
- f - Frequency of the radiation (Hz)

Alternatively, since frequency f and wavelength λ are related by the speed of light c ,

$f = \frac{c}{\lambda}$, the relation can also be written as:

$$E = \frac{hc}{\lambda} \quad (2)$$

where:

- λ - Wavelength of the radiation (m)
- c - Speed of light (3×10^8 m/s)

LEDs (light-emitting diodes) emit different colors of light, with each corresponding to a specific wavelength determined by the properties of the material used. These materials are semiconductors with specific energy band gaps (E_g), which represent the energy difference between the valence band (where electrons are bound) and the conduction band (where electrons are free to move) (Balkan & Erol, 2021). During the process of electroluminescence, when current flows through the LEDs, electrons recombine with holes in the semiconductor material. Within this recombination, the electrons transition to a lower energy state, releasing energy in the form of photons. The energy of each emitted photon is approximately equal to the E_g of the semiconductor material used (Petit et al., 2016). According to equation (2), this energy is inversely proportional to the wavelength of the emitted light—meaning that higher-energy photons correspond to shorter wavelengths.

In LEDs, due to the characteristics of semiconductor diodes, current is allowed to flow in only one direction and light is emitted only after reaching a certain threshold voltage. This voltage provides the minimum energy required to excite electrons across the band gap and corresponds to the band gap energy of the material. Assuming that each electron passing through the LED recombines and emits a single photon, the energy provided by the voltage can be equated to the energy of the emitted photon, with the equation:

$$E = eV \quad (3)$$

where:

- V - Threshold voltage (V)
- e - Elementary charge (1.60×10^{-19} C)

Thus, at the moment the LED begins to emit light, the applied voltage reaches the threshold voltage,

$$E = \frac{hc}{\lambda} = eV \quad (4)$$

Rearranging the equation and substituting V_T for threshold voltage V :

$$V_T = \frac{hc}{\lambda e} = \frac{1}{\lambda} \times \frac{hc}{e} \quad (5)$$

Since $\frac{hc}{e}$ is constant, it can be conducted that $V_T \sim \frac{1}{\lambda}$ or V_T is inversely proportional to λ .

According to this equation, it can be hypothesized that as the wavelength λ decreases, the threshold voltage (V_T) increases. Therefore, LEDs emitting shorter wavelengths (e.g., violet light) are expected to require higher threshold voltages, while those emitting longer wavelengths (e.g., red light) will require lower threshold voltages.

To investigate this relationship, the experiment will utilize a potential divider circuit and LEDs of seven different colors with known wavelengths to measure their corresponding threshold voltages. Subsequently, the experimental data will be used to determine the value of Planck's constant.

Parameters

Variables		Explanation	Apparatus
Independent Variable	Wavelength of the LEDs (λ)	Since the energy of emitted photons depends on the wavelength of the light, using LEDs of different colors with corresponding wavelengths results in varying photon energies and influences the threshold voltage recorded for each LED to emit light, allowing investigation of the relationship between these two variables.	Seven groups of LEDs in different colors with known wavelengths, as labeled by the manufacturer, will be used, ranging from longer wavelengths (red) to shorter wavelengths (violet).
Dependent Variable	Threshold Voltage (V_T)	Threshold voltage is the minimum voltage at which an LED begins to emit visible light. This value corresponds to the energy required for electrons to cross the band gap and emit photons.	The threshold voltage will be measured using a multimeter (± 0.01 V) while gradually increasing the voltage until the LED just begins to emit visible light.

Table 1: Independent and Dependent Variables

Controlled Variables		
Variables	Significance	Means of control
LED characteristics	Variations in LED type, size, or material may affect the threshold voltage and light emission behavior.	Use LEDs of the same type and brand, only the color (wavelength) is different.
Current through the circuit	Excessive current will damage the LED and affect the accuracy of the threshold voltage measurement.	A light bulb is connected in series to limit the current flow and prevent overcurrent.
Measurement method	The use of inconsistent measuring equipment can cause differences in voltage readings.	The same multimeter was used throughout the experiment to maintain consistent and accurate measurements.
Temperature	Fluctuations in temperature can affect the electrical properties of LEDs, therefore affecting the measurements of threshold voltages.	The experiment will be carried out at room temperature in a relatively stable environment without heat source.
Ambient light conditions	External light sources can interfere with the observation of when the LED begins to emit light.	The experiment will be conducted in a relatively dark room to minimize interference from outside light sources.

Table 2: Controlled Variables

Uncontrolled Variables	
Variables	Significance
Manufacturing Variations in LEDs	Even LEDs of the same color from the same manufacturer may have slightly different band gap energies, which can affect the threshold voltage.
Photonic Emission Inefficiencies	Due to the inefficiency of LED materials, not all electrons can be recombined to emit photons. Some of the energy may be lost in the form of heat or other non-radiative processes, resulting in slight deviations from the expected results.

Table 3: Uncontrolled Variables

Apparatus

- 7 groups of LEDs, each emitting a different color, with known wavelengths labeled by the manufacturer (± 5 nm)
- Variable DC power supply (0–12 V, max current 2 A)
- Multimeter (± 0.01 V)
- Light bulb, used as a current-limiting resistor
- Connecting wires and alligator clips, used to connect components in the circuit
- Self-built Variable Resistor - resistance wire mounted on a ruler

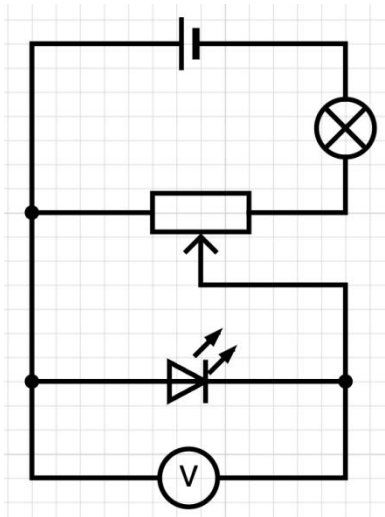


Figure 1. Diagram of experimental setup

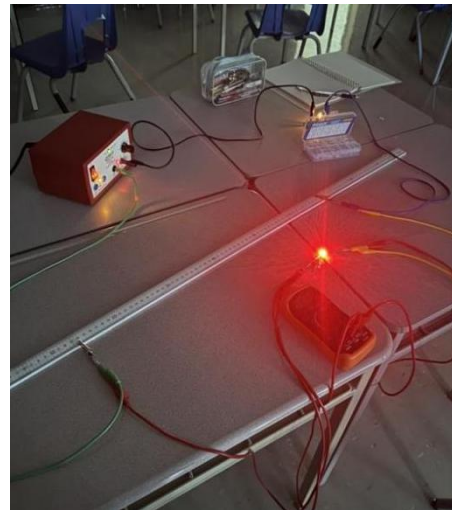


Figure 2. Picture of experimental setup

Procedure

1. Set up the circuit as shown in the diagram. The power supply is set to 3 V. Connect a light bulb in series to limit the current in the circuit, and a multimeter in parallel with the LED to measure its voltage. A sliding contact (movable crocodile clip) is used along the resistance wire to vary the effective resistance and control the voltage across the LED.
2. Use crocodile clips to connect an LED into the circuit. Ensure the LED is forward-biased so that it can emit light when the threshold voltage is reached.
3. Position the sliding contact near the start of the resistance wire, where the resistance is lowest. This ensures the voltage across the LED is initially below its threshold.
4. Slowly move the sliding contact along the resistance wire to increase the effective resistance in the circuit, thereby gradually increasing the voltage across the LED.
5. Observe the LED carefully and stop adjusting the slider as soon as it begins to emit visible light.
6. Record the voltage displayed on the multimeter, which represents the threshold voltage of the tested LED.
7. Replace the previous LED with 6 other LEDs of the same color, and repeat steps 2–6 for each to obtain a full data set for that wavelength. This helps avoid the risk of inaccurate data for the entire group caused by relying on a single LED that may have a manufacturing defect.
8. Replace the current set of LEDs with LEDs of a different color, and repeat steps 2–7 to collect data for the new wavelength.
9. Repeat step 8 until threshold voltage data has been collected for all seven wavelengths of LEDs.

Preliminary Trials

Preliminary tests are conducted to assess the feasibility of the experiment and optimize the experimental setup. Initially, the experiment used a sliding rheostat for voltage adjustment, but it was found unable to fine-tune the resistance and accurately determine the threshold voltage. Hence, the setup was modified to use a self-made variable resistor consisting of a 1-meter constantan wire (SWG 30) mounted on a ruler. This enables improved voltage

control and more accurate measurements.

Environmental Considerations

To prevent improper disposal of electronic components, such as damaged and unusable LEDs from the experiment, to cause harmful chemicals (e.g., heavy metals or plastics) to leach into the environment, all unusable electronic parts should be recycled through appropriate e-waste recycling programs. Meanwhile, all components should be handled carefully to avoid damage. Furthermore, minimizing the use of electrical energy during the experiment, such as turning off the power when not in use, can help reduce the environmental impact.

Ethical Considerations

This experiment does not involve ethical concerns, as no living organisms are involved in the data collection process.

Safety Considerations

While handling the electrical circuit, the power supply is ensured to turn off during adjustments to avoid shocks. Meanwhile, a light bulb is used in series to limit current and prevent damage to the LED. Long periods of operation at high current are avoided to prevent overheating of the resistance wires and other components, ensuring safe and efficient functioning of the circuit. Hands are kept dry when handling equipment, and cables are arranged neatly to avoid tangling or tripping. Components are positioned securely to maintain a clear and stable working area.

Raw Data

Color	Wavelength (λ) [nm] $\pm 5\text{ nm}$	Threshold Voltage Trials $\pm 0.02\text{ V}$							Mean Threshold Voltage (V_T) [V]	Uncertainty of Mean Threshold Voltage (ΔV_T) [V]
		Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7		
Red	650	1.48	1.49	1.46	1.47	1.48	1.49	1.48	1.48	0.015
Orange	610	1.53	1.54	1.55	1.56	1.53	1.52	1.56	1.54	0.020
Yellow	580	1.64	1.66	1.65	1.63	1.64	1.63	1.65	1.64	0.015
Green	540	1.86	1.87	1.89	1.88	1.88	1.89	1.87	1.88	0.015
Blue	500	2.02	2.01	2.00	1.99	2.03	2.00	2.02	2.01	0.020
Indigo	470	2.21	2.23	2.20	2.19	2.22	2.21	2.20	2.21	0.020
Violet	430	2.45	2.42	2.43	2.44	2.43	2.41	2.44	2.43	0.020

Table 4: Raw Data – Threshold Voltage Measurements and their Uncertainties

Although the uncertainty has already been estimated as $\pm 0.02\text{ V}$ based on the 20 V range of the multimeter, the uncertainties for mean threshold voltage are still calculated since some of them are smaller, but the majority aligns with estimates.

To obtain these data, the following calculations has been processed, $\lambda = 540\text{ nm}$ is used as an example.

Calculation of mean threshold voltage V_T :

$$V_T = \frac{V_1 + V_2 + \dots + V_7}{7}$$

$$V_T = \frac{1.86 + 1.87 + 1.89 + 1.88 + 1.88 + 1.89 + 1.87}{7} \approx 1.88\text{ V}$$

Calculation of uncertainty ΔV_T :

$$\Delta V_T = \frac{V_{\max} - V_{\min}}{2}$$

$$\Delta V_T = \frac{1.89 - 1.86}{2} = 0.015\text{ V}$$

The data is plotted on a $V_T - \lambda$ graph:

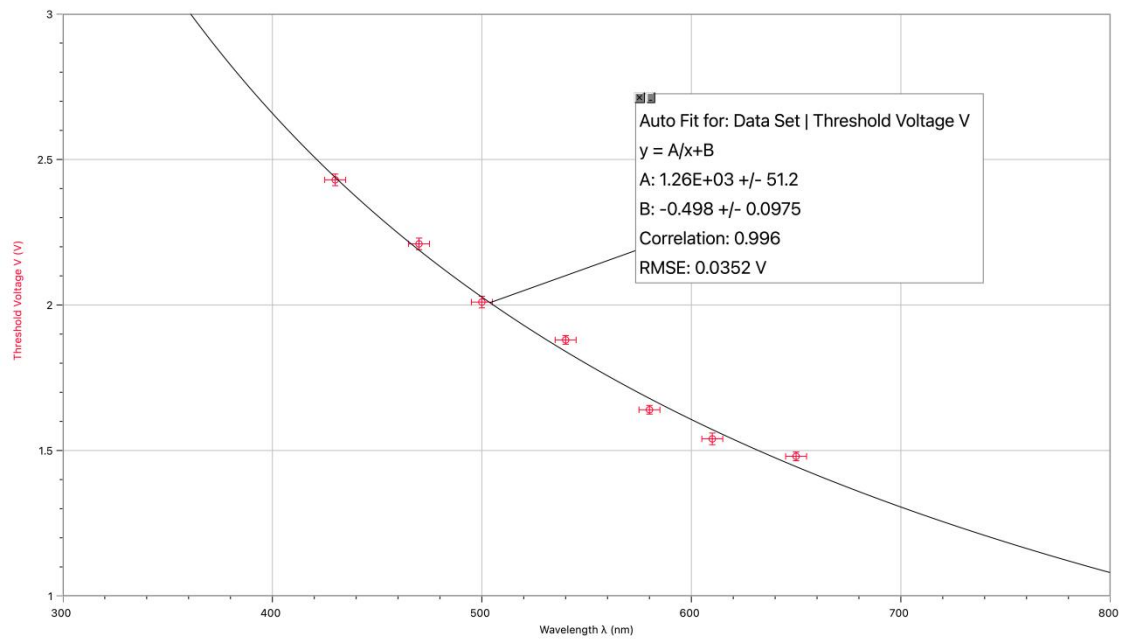


Figure 3. $V_T - \lambda$ graph with experimental data

The high correlation ($R^2 = 0.996$) and low RMSE (0.0352 V) indicate the experimental data closely follow the previous hypothesis. Although some data points do not lie exactly on the curve, they are very close to it. This deviation of these points may suggest the presence of random errors in the experiment.

Processed Data

Mean Threshold Voltage (V_T) [V]	Wavelength (λ) [nm] $\pm 5nm$	Inverse Wavelength ($\frac{1}{\lambda}$) [m^{-1}]	Uncertainty of Inverse Wavelength ($\Delta\frac{1}{\lambda}$) [m^{-1}]
1.48 ± 0.015	650	1.54×10^6	11800
1.54 ± 0.020	610	1.64×10^6	13400
1.64 ± 0.015	580	1.72×10^6	14900
1.88 ± 0.015	540	1.85×10^6	17100
2.01 ± 0.020	500	2.00×10^6	20000
2.21 ± 0.020	470	2.13×10^6	22600
2.43 ± 0.020	430	2.33×10^6	27000

Table 5: Processed Data – Inverse Wavelength Calculations and their Uncertainties

To process these data, following calculations are used, taking $\lambda = 540 \text{ nm}$ as an example.

Calculation of inverse wavelength $\frac{1}{\lambda}$:

$$\frac{1}{\lambda} = \frac{1}{540} = 0.00185185 \text{ nm}^{-1} \approx 1.85 \times 10^6 \text{ m}^{-1}$$

Calculation of uncertainty $\Delta \frac{1}{\lambda}$:

$$\Delta \frac{1}{\lambda} = \frac{1}{\lambda} \times \frac{\Delta \lambda}{\lambda} = \frac{\Delta \lambda}{\lambda^2}$$

$$\Delta \frac{1}{\lambda} = \frac{5}{540^2} \approx 1.71 \times 10^{-5} \text{ nm}^{-1} \approx 17100 \text{ m}^{-1}$$

The data is plotted on a $V_T - \frac{1}{\lambda}$ graph:

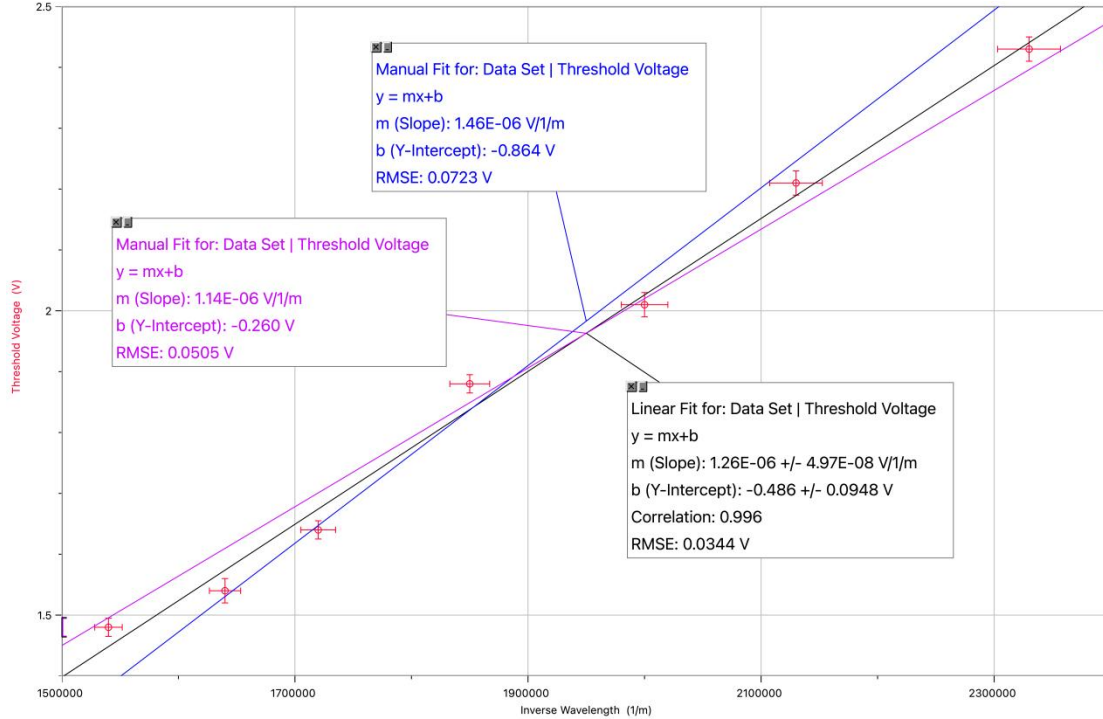


Figure 4. $V_T - \frac{1}{\lambda}$ graph with best (black), maximum (blue), and minimum (violet) fit lines

These two variables, when plotted, produced a straight line, indicating the linear relationship between threshold voltage and inverse wavelength, which follows results of the formula derivation before. The high correlation ($R^2 = 0.996$) and low RMSE (0.0344 V) for the best linear fit also suggests the strong positive correlation between the two variables. As before, the error bars are fairly small, and there are still several data points that are close but not touching the best fit line, suggesting the presence of random errors in the experiment.

As shown in the graph, the best fit line has a slope of $m = 1.26 \times 10^{-6} \text{ Vm}$ with a y-intercept $b = -0.486 \text{ V}$. The maximum fit and minimum fit line have $m_{max} = 1.46 \times 10^{-6} \text{ Vm}$, $b = -0.864 \text{ V}$ and $m_{min} = 1.14 \times 10^{-6} \text{ Vm}$, $b = -0.260 \text{ V}$ respectively.

Following calculations are processed to determine the uncertainty of slope m and y-intercept values:

Calculation of uncertainty of the slope m :

$$\Delta m = \frac{m_{max} - m_{min}}{2}$$

$$\Delta m = \frac{1.46 \times 10^{-6} - 1.14 \times 10^{-6}}{2} = 0.16 \times 10^{-6} \text{ Vm}$$

Calculation of uncertainty of y-intercept b :

$$\Delta b = \frac{b_{max} - b_{min}}{2}$$

$$\Delta b = \frac{-0.260 - (-0.864)}{2} = 0.302 \text{ V}$$

Therefore, the value of slope m is $(1.26 \pm 0.16) \times 10^{-6} \text{ Vm}$, and the value of y-intercept b is $-0.486 \pm 0.302 \text{ V}$.

In term of m , the relationship between threshold voltage and wavelength can be mathematically arranged as:

$$V_T = m \times \frac{1}{\lambda} + b = \frac{1.26 \times 10^{-6} \text{ Vm}}{\lambda} - 0.486 \text{ V}$$

Using equation (5), $V_T = \frac{hc}{\lambda e}$, it can be obtained that,

$$m = \frac{h \times c}{e} - b \times \lambda$$

Since the value of $b \times \lambda$ is very close to 0, it can be considered negligible in the calculation, therefore, Planck's constant can be calculated by solving for h :

$$h = \frac{m \times e}{c} = \frac{1.26 \times 10^{-6} \times 1.60 \times 10^{-19}}{3.00 \times 10^8} = 6.72 \times 10^{-34} \text{ Js}$$

The uncertainty of h is calculated:

$$\Delta h = h \left(\frac{\Delta m}{m} \right) = 6.72 \times 10^{-34} \times \frac{0.16 \times 10^{-6}}{1.26 \times 10^{-6}} = 8.53 \times 10^{-35} \text{ Js}$$

The percentage uncertainty:

$$\frac{\Delta h}{h} = \frac{8.53 \times 10^{-35}}{6.72 \times 10^{-34}} \approx 12.7 \%$$

In this investigation, Planck's constant has been found to be $h = (6.72 \pm 0.85) \times 10^{-34}$ Js.

Conclusion

While answering the question, “**How can Planck's constant (h) be determined by investigating the relationship between the threshold voltage (V) and wavelength (λ) of light emitted by different colors of light-emitting diodes?** ”, a strong correlation between these two variables was observed, consistent with the hypothesis that V_T is inversely proportional to λ .

The best-fit line produced a slope of $m = 1.26 \times 10^{-6}$ Vm, enabling the calculation of Planck's constant as $h = (6.72 \pm 0.85) \times 10^{-34}$ Js with a percentage uncertainty of 12.7%. The high correlation coefficient and low RMSE in both graphs confirm the reliability of the experiment results.

Despite some minor deviation of data points from the best-fit line, it is likely due to the random errors such as manufacturing variations in the LEDs, the results are consistent with the theoretical model in general. However, the value obtained for Planck's constant is not perfectly in line with the currently accepted value of $h = 6.62607015 \times 10^{-34}$ Js (Britannica, 2025), it differs by 1.42%.

$$\% \text{ Error} = \frac{\text{Experimental Value} - \text{Accepted Value}}{\text{Accepted Value}} \times 100\%$$

$$\% \text{ Error} = \frac{6.72 \times 10^{-34} - 6.62607015 \times 10^{-34}}{6.62607015 \times 10^{-34}} \approx 1.42 \%$$

The percentage error between the obtained Planck's Constant and the accepted standard value is well below 5%, this indicates that the experimental results are highly accurate and provide strong support for the proposed hypothesis. However, the slight overestimation of the obtained Planck's constant may suggest the presence of a systematic error in the experimental procedure that led to consistent larger measurements of the threshold voltage. Despite this, the close agreement between the experimental and theoretical values confirms the overall validity of the methodology used and demonstrates that this investigation provides a reliable approach to estimate Planck's constant.

Evaluation

Strengths

Strengths	Discussion
Multiple Trials for Accuracy	Seven trials are conducted for each wavelength of LED to ensure the data collected is consistent and the impact of random errors is minimized .
Change of LED for Each Trial	After each trial, the LED is replaced with a new one to reduce the risk of errors caused by wear, overheating, or inconsistent band gap energy of single LEDs. The results represent the general behavior of a specific wavelength, rather than individual variations.
Comprehensive Data Collection	Seven different colors, covering the entire spectrum from red to purple, make the investigation of this relationship boarder and the conclusions more valid.
Controlled Variables	Apart from the replacement of LEDs, all devices used in the experiment remained consistent, and factors such as temperature and measurement methods were strictly controlled, therefore minimizing systematic errors and enhancing the reliability of experimental data.
Experimental Setup	The self-built variable resistor provided precise control over voltage adjustments, while the relatively dark environment allowed for more accurate observation of threshold voltage. Additionally, the small light bulb in the circuit protected the LED from overcurrent damage.
Replicability	The experiment used accessible and standardized components, such as commercially available LEDs and multimeter. It is easy to replicate and verify the results.

Table 6: Strengths and Discussion

Limitations and Improvements

Source of error	Impact on the results	Improvements
Random Errors		
Visual Observation	Determining the threshold voltage relies on visually identifying the point at which the LED begins to emit light. This is subjective and could lead to overestimated voltage values due to human observational delay.	Use a photodiode or light sensor placed in front of the LED to detect the exact moment light is emitted. These sensors offer greater precision than the human eye and can be connected to a multimeter for accurate detection.
Manual Adjustments of Resistance	Manually adjusting resistors makes it difficult to finely control the voltage, resulting in inconsistent threshold detection and greater variation between trials.	Use a precision potentiometer or a digital programmable resistor module for finer adjustments.
Contact Resistance	Contact points (e.g., crocodile clips and wires) may introduce extra resistance, especially if corroded or oxidized, which can result in altered voltage measurements.	Use high-quality, low-resistance connectors such as gold-plated banana plugs, and regularly check for corrosion to ensure accurate and stable voltage readings.

Table 7: Random Errors and Improvements

Source of error	Impact on the results	Improvements
Systematic Errors		
Light Bulb in Circuit	The light emitted by the bulb may interfere with the observer in accurately detecting the moment LED begins to emit light, especially in dim conditions.	Enclose the bulb in a dark box or use a high-wattage resistor instead to limit current without introducing additional light.
Wavelength Inaccuracy	Manufacturer-specified LED wavelengths are approximate and may not reflect the actual emitted wavelength due to batch variations. This introduces errors when processing the data.	Use a spectrometer to measure the actual peak emission wavelength of each LED in the experiment.
Voltage Resolution	The multimeter's limited resolution may miss small changes near the V_T , reducing precision in determining the exact point of light emission.	Use a high-resolution digital multimeter (± 0.001 V) or connect the circuit to a data logger.
Energy Losses in LED	Some of the energy in the LED may be lost as heat or through non-radiative recombination, resulting in a higher threshold voltage measured.	Account for these losses in theoretical models or use LEDs with higher quantum efficiency.

Table 8: Systematic Errors and Improvements

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